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Presentation Report

on

<u>LUCAS – KANADE ALGORITHM</u>

for

OPTICAL FLOW

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LUCAS–KANADE ALGORITHM: Optical Flow and Motion Estimation

1. Introduction

Motion estimation is a foundational technique in computer vision used to determine the movement of objects between consecutive frames in a video. One of the most effective and widely used methods for estimating this motion is the Lucas–Kanade Algorithm, which belongs to the class of optical flow algorithms.

Optical flow refers to the apparent motion of brightness patterns in a sequence of images. The Lucas–Kanade method is particularly well-suited for small, consistent movements and is often used in real-time tracking applications.

2. Optical Flow Fundamentals

Optical Flow is a vector field where each vector denotes the apparent motion of brightness patterns from one frame to another. It estimates how pixels shift over time due to the motion of objects or the camera.

Key Applications

- Video compression
- Object tracking
- Motion-based segmentation
- Robot navigation
- Augmented reality

3. Fundamental Assumptions in Optical Flow

Assumption #1: Brightness Constancy

The intensity of a pixel remains constant between frames:

$$I(x,y,t)=I(x+\delta x,y+\delta y,t+\delta t)$$

This implies that the object moves, but the observed brightness of any point remains unchanged.

Assumption 2: Small Motion

The displacement of pixels between frames is small:

$$u(x,y),v(x,y)\approx 0$$

This assumption allows us to linearize the motion equations using Taylor series.

Taylor Series Expansion

To approximate changes in intensity, the Taylor series is used:

$$I\ (x+\delta x,y+\delta y,t+\delta t) \approx I(x,y,t)+Ixu+Iyv+It$$

Where:

- Ix,IyI_x, I_yIx,Iy: spatial gradients
- ItI_tIt: temporal gradient
- u,vu, vu,v: optical flow components

Substituting and rearranging yields the Optical Flow Constraint Equation:

$$Iu+Iyv+It=0$$

Lucas–Kanade Method

Spatial Coherence Constraint

The Lucas–Kanade method assumes that the flow is essentially constant within a small neighborhood. Thus, instead of computing flow for each pixel individually, it solves a system of equations over a window of pixels.

This leads to the overdetermined system:

$$Av=B$$

Where:

- A contains spatial gradients Ix and Iy
- B=−It

Solving by **Least Squares**:

$$B=(A^TA)^{-1}A^Tb$$

SOLVING THE EQUATION:

The Lucas–Kanade method assumes that the displacement of the image contents between two nearby instants (frames) is small and approximately constant within a neighborhood of the point p under consideration. Thus the optical flow equation can be assumed to hold for all pixels within a window centered at p. Namely, the local image flow (velocity) vector (V_x, V_y) must satisfy

$$\begin{aligned} I_x(q_1)V_x + I_y(q_1)V_y &= -I_t(q_1) \\ I_x(q_2)V_x + I_y(q_2)V_y &= -I_t(q_2) \\ &\vdots \\ I_x(q_n)V_x + I_y(q_n)V_y &= -I_t(q_n) \end{aligned}$$

where $q_1, q_2, \dots, q_n$ are the pixels inside the window, and $I_x(q_i), I_y(q_i), I_t(q_i)$ are the partial derivatives of the image I with respect to position x, y and time t, evaluated at the point q_i and at the current time.

$$A = \begin{bmatrix} I_x(q_1) & I_y(q_1) \\ I_x(q_2) & I_y(q_2) \\ \vdots & \vdots \\ I_x(q_n) & I_y(q_n) \end{bmatrix} \qquad v = \begin{bmatrix} V_x \\ V_y \end{bmatrix} \qquad b = \begin{bmatrix} -I_t(q_1) \\ -I_t(q_2) \\ \vdots \\ -I_t(q_n) \end{bmatrix}$$

$$\begin{bmatrix} V_x \\ V_y \end{bmatrix} = \begin{bmatrix} \sum_i I_x(q_i)^2 & \sum_i I_x(q_i)I_y(q_i) \\ \sum_i I_y(q_i)I_x(q_i) & \sum_i I_y(q_i)^2 \end{bmatrix}^{-1} \begin{bmatrix} -\sum_i I_x(q_i)I_t(q_i) \\ -\sum_i I_y(q_i)I_t(q_i) \end{bmatrix}$$

Limitations of Lucas–Kanade

- **Aperture Problem**: Ambiguity in determining motion from local information
- **Requires Good Texture**: Needs high gradients in both directions (corner detection)
- **Small Motion Assumption**: Large movements break the algorithm
- **Matrix Inversion**: Requires $A^TA$to be invertible and well-conditioned

ANALYSING M (A^TA) MATRIX

Eigenvalue conditions:

- $\lambda 1, \lambda 2 > 0$
- $\lambda 1 / \lambda 2 \approx 1$

Improvements and Extensions

- **Robust Statistics**: Reduce the influence of outliers using weighting or RANSAC
- **Multi-Resolution Analysis**: Apply pyramid representations to detect larger motions
- **KLT Tracker**: Kanade–Lucas–Tomasi algorithm improves tracking with feature detection and scale adaptation
- **Refinement Techniques**: Use previous estimates or coarse-to-fine strategies

Conclusion

The Lucas–Kanade algorithm is a powerful technique for estimating motion in image sequences. Its simplicity, combined with reasonable accuracy for small motions, makes it an essential tool in computer vision. Despite limitations, enhancements such as pyramidal approaches and robust statistics have broadened its applicability.

References

- Lucas, B. D., & Kanade, T. (1981). An iterative image registration technique with an application to stereo vision.
- Horn, B. K. P., & Schunck, B. G. (1981). Determining optical flow.